Exp6

Server

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <arpa/inet.h>
#define PORT 8080
#define MAX 1024
void translate(char *str)
{
    char temp[MAX];
    char result[MAX] = "";
    char *token;
    strcpy(temp, str);
    token = strtok(temp, " ,.!?\n");
    while (token != NULL)
    {
        char word[50];
        strcpy(word, token);
        // Convert token to lowercase
        for (int i = 0; word[i]; i++)
            word[i] = tolower(word[i]);
        if (strcmp(word, "tbh") == 0)
            strcat(result, "to be honest");
        else if (strcmp(word, "idc") == 0)
            strcat(result, "I don't care");
        else if (strcmp(word, "ig") == 0)
            strcat(result, "I guess");
        else if (strcmp(word, "tbf") == 0)
            strcat(result, "to be fair");
        else if (strcmp(word, "atm") == 0)
            strcat(result, "at the moment");
        else if (strcmp(word, "irl") == 0)
            strcat(result, "in real life");
        else if (strcmp(word, "lol") == 0)
            strcat(result, "laughing out loud");
        else if (strcmp(word, "asap") == 0)
            strcat(result, "as soon as possible");
```

```
        else if (strcmp(word, "omg") == 0)
            strcat(result, "oh my God");
        else if (strcmp(word, "ttyl") == 0)
            strcat(result, "talk to you later");
        else if (strcmp(word, "idk") == 0)
            strcat(result, "I don't know");
        else if (strcmp(word, "nvm") == 0)
            strcat(result, "never mind");
        else
            strcat(result, token);
        strcat(result, " ");
        token = strtok(NULL, " ,.!?\n");
    }
    strcpy(str, result);
}
int main()
{
    int sockfd;
    char buffer[MAX];
    struct sockaddr_in server_addr, client_addr;
    socklen_t len = sizeof(client_addr);
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(sockfd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("UDP Server Running...\n");
    recvfrom(sockfd, buffer, MAX, 0,(struct sockaddr *)&client_addr, &len);
    printf("Received: %s\n", buffer);
    translate(buffer);
    sendto(sockfd, buffer, strlen(buffer) + 1, 0,
           (struct sockaddr *)&client_addr, len);
    printf("Translated message sent.\n");
    close(sockfd);
    return 0;
}
```

Client

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#define PORT 8080
#define MAX 1024
int main()
{
    int sockfd;
    char message[MAX];
    char buffer[MAX];
    struct sockaddr_in server_addr;
    socklen_t len = sizeof(server_addr);
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");
    printf("Enter sentence:\n");
    fgets(message, MAX, stdin);
    sendto(sockfd, message, strlen(message) + 1, 0,(struct sockaddr *)&server_addr, len);
    recvfrom(sockfd, buffer, MAX, 0,(struct sockaddr *)&server_addr, &len);
    printf("\nTranslated Sentence:\n%s\n", buffer);
    close(sockfd);
    return 0;
}
```

Server

```
UDP Server Running...
Received: TBH, idk what to do ATM, LOL.

Translated message sent.
```

Client

```
Enter sentence:
TBH, idk what to do ATM, LOL.

Translated Sentence:
to be honest I don't know what to do at the moment laughing out loud
```